AI + IOT · ENVIRONMENTAL MONITORING

AI-Powered Multi-Sensor Environmental Monitoring and Prediction System

Four sensors on an ESP32, with machine learning that predicts air quality and flags unusual readings

DHT22 | MQ135 | IR sensor | Ultrasonic | ESP32 + Blynk | Random Forest | Isolation Forest

THE SENSORS

Four sensors, four kinds of data

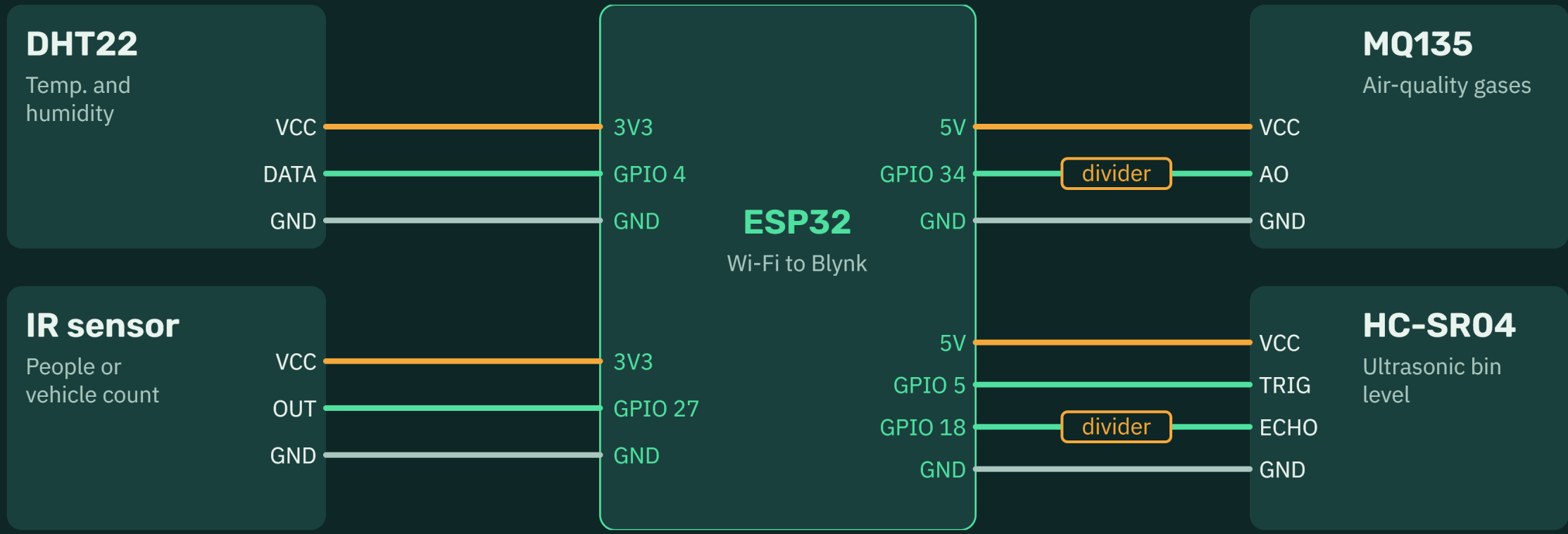
Sensor	Measures	Role in the AI system
DHT22	Temperature, humidity	Weather context for the model
MQ135	Air-quality gases	The pollution level to predict
IR sensor	People or vehicles passing	Activity that can drive pollution
Ultrasonic	Distance to bin contents	Bin fill level as an odour signal

Weather, pollution, activity and waste level together give the model far more context than one gas sensor alone.

The IR and ultrasonic roles are a design choice (a people counter and a bin-level gauge) and can be swapped for your site.

HARDWARE WIRING

Circuit: four sensors on one ESP32



Amber: power · Green: signal · Grey: ground · Dividers keep the 5 V MQ135 and ECHO signals under 3.3 V

WORKING SYSTEM

How the system works end to end

1 • Sense
DHT22, MQ135, IR and ultrasonic readings

2 • ESP32
Reads, averages and sends over Wi-Fi

3 • Blynk
Stores and shows live values

4 • AI model
Predicts and flags anomalies

5 • Dashboard
Forecast, flags and alerts

On the ESP32

Read the four sensors, smooth out noise, upload.

on pc

Build features, run both models, write results back to Blynk.

AI STEP 1 · PREPARE THE DATA

From raw readings to model inputs

Raw reading	Model input	Why it helps
MQ135 voltage	Level, 1-hour change	Shows if air is worsening
DHT22	Temperature, humidity	Weather shapes gas build-up
IR sensor	Count per hour	Measures activity
Ultrasonic	Bin fill %, rise per hour	Tracks waste build-up
Clock	Hour of day	Captures daily cycles

Each hour becomes one row of numbers. That table is what the model learns from.

AI STEP 2 · LEARN FROM THE PAST

Learning from past readings

Every row is a worked example: the conditions at one moment, and what MQ135 read an hour later.

Time	Temp °C	Humidity %	MQ135	People/h	Bin fill %	Answer: +1 h
07:00	24	78	52	120	30	71
09:00	27	70	71	310	38	84
13:00	33	45	60	180	52	55
18:00	30	58	66	260	61	78
22:00	25	75	58	40	70	63

The first six columns are inputs and the last is the answer. Training on weeks of rows teaches the model which combinations lead to higher or lower readings later.

AI STEP 3 · RANDOM FOREST

From new readings to a forecast

New reading

31 °C, 62 % humidity

MQ135 at 76, rising

260 people per hour

Bin 61 % full

Each tree votes

Tree 1 predicts 80

Tree 2 predicts 85

Tree 3 predicts 81

A real forest has 100 or more

Average

82

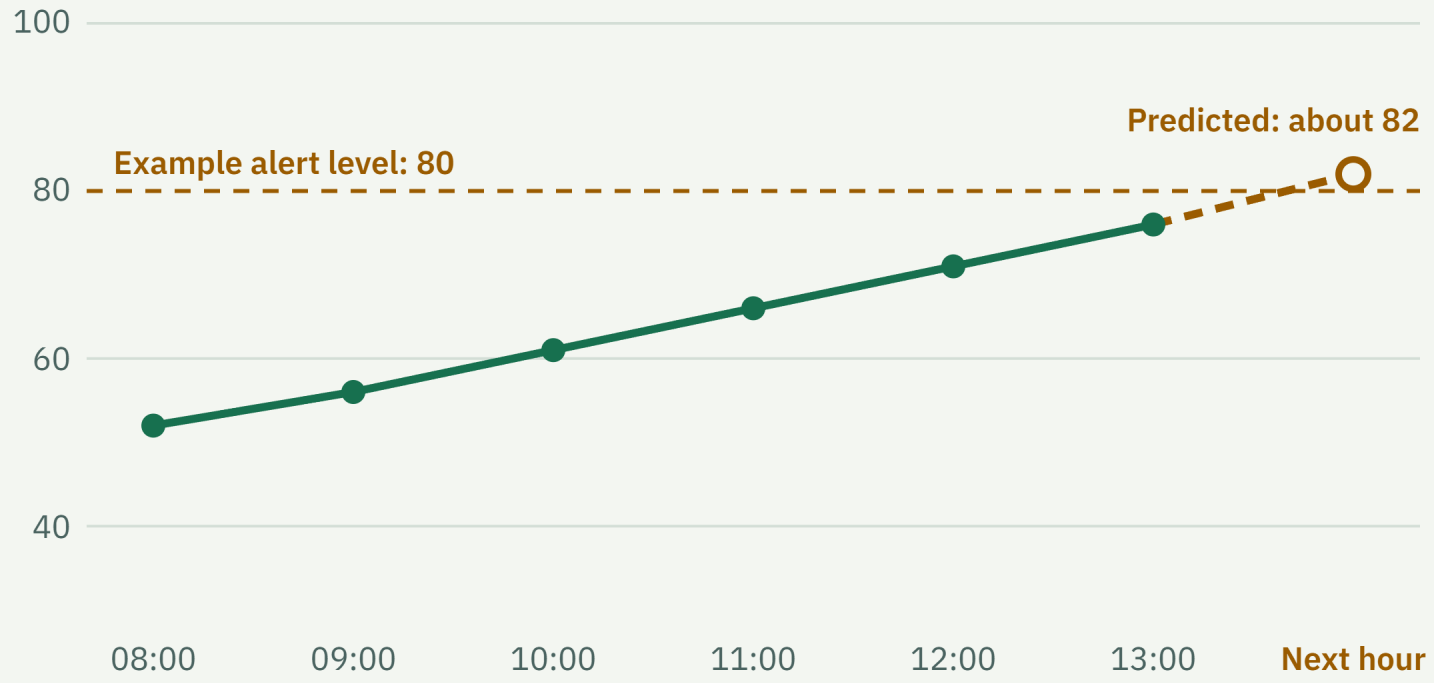
Predicted MQ135 in 1 hour

Why a forest: averaging many trees stops one odd tree from misleading the result.

Trust check: test on days the model never saw and report the average error.

AI STEP 4 · PREDICT FROM PATTERNS

A rising trend becomes an early warning



Solid line: logged MQ135 · **Dashed: model prediction** · Values are illustrative

Pattern

Readings climbed steadily for five hours as activity grew.

Prediction

The model expects about 82 in the next hour.

Action

That crosses the example limit of 80, so Blynk warns early.

AI STEP 5 · ISOLATION FOREST

Spotting unusual readings across all sensors

The model learns what normal combinations look like. A reading that is easy to isolate from the rest of the history gets flagged.

Case	MQ135	People/h	Bin fill %	Result
Normal	66	260	61	Normal: no flag
Gas spike	180	240	61	Flag: smoke, dust or fault?
Bin jump	70	250	5 (was 61)	Flag: emptied or glitch?

A flag asks for verification. It does not declare that pollution has risen.

OUTPUT AND LIMITS

What the system does with the AI output

Forecast	Anomaly flag	Action on Blynk
Below limit	No	Show the forecast
Above limit	No	Send an early warning
Below limit	Yes	Mark the reading as "check sensor"
Above limit	Yes	Warn and ask for an on-site check

Limits to state openly

MQ135 gives a gas index, not PM2.5. The model needs weeks of your own data and must be tested on unseen days.

WRAP-UP

Conclusion and results

Conclusion

Four sensors give weather, pollution, activity and waste context.

Random Forest turns past data into a one-hour forecast.

Isolation Forest flags readings that need a check.

Sensors and Blynk make it IoT. The trained models make it AI.

Results to report

Measure	Your result
Forecast error (MAE)	[__]
Anomalies flagged	[__ of __]
Early warnings sent	[__]
Warning lead time	[__ min]

Fill these in from your own test run on days the model never saw.